

Physics 829: Homework Set No. 2

**Due date: Friday, April 13, 2012, 1:00pm
in PRB M2043 (Biao Huang's office)**

Total point value of set: 60 points

Problem 1 (10 pts.): Exercise 17.2.1 (Shankar, p. 457)

Problem 2 (10 pts.): Exercise 17.2.2 (Shankar, p. 457)

Problem 3 (10 pts.): Exercise 17.2.3 (Shankar, p. 457)

Problem 4 (10 pts.): Exercise 17.2.4 (Shankar, p. 457)

Problem 5 (10 pts.): Compute $|n^2\rangle$ and E_n^3 for a harmonic oscillator in a constant external electric field.

Problem 6 (10 pts.): Exercise 17.2.7 (Shankar, p. 463)

Exercise 17.2.1, p. 457

$$(1) \quad \hat{H} = \hat{H}^0 + \hat{H}^1 = \frac{\hat{p}^2}{2m} + \frac{m}{2}\omega^2 \hat{x}^2 + \lambda \hat{x}^4$$

$$\hat{H}^0 |n^0\rangle = E_n^0 |n^0\rangle = (n + \frac{1}{2})\hbar\omega |n^0\rangle$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}^+ + \hat{a}) \Rightarrow \hat{x}^4 = \left(\frac{\hbar}{2m\omega}\right)^2 (\hat{a}^{+4} + \dots + \hat{a}^4)$$

For the 1st order energy shift

$$E_n^1 = \langle n^0 | \lambda \hat{x}^4 | n^0 \rangle$$

We need only those terms in $\lambda \hat{x}^4$ with equal numbers of $\hat{a}$ and $\hat{a}^+$ operators; there are 6 of them.

$$\begin{aligned} (\hat{a}^+ + \hat{a})^4 &= (\hat{a}^{+2} + \hat{a}^+ \hat{a} + \hat{a} \hat{a}^+ + \hat{a}^2)^2 = \quad (\hat{a} \hat{a}^+ = 1 + \hat{a}^+ \hat{a} = 1 + \hat{n}) \\ &= (\hat{a}^{+2} + 2\hat{a}^+ \hat{a} + 1 + \hat{a}^2)^2 \\ &= (\dots + \hat{a}^{+2} \hat{a}^2 + (2\hat{a}^+ \hat{a} + 1)^2 + \hat{a}^2 \hat{a}^{+2} + \dots) \end{aligned}$$

I listed only the interesting terms that contribute to the expectation value. Using $\hat{a} \hat{a}^+ = 1 + \hat{a}^+ \hat{a}$ repeatedly we get

$$\hat{a}^{+2} \hat{a}^2 = \hat{a}^+ (\hat{a} \hat{a}^+ - 1) \hat{a} = \hat{n}^2 - \hat{n}$$

$$\begin{aligned} \hat{a}^2 \hat{a}^{+2} &= \hat{a} (\hat{a}^+ \hat{a} + 1) \hat{a}^+ = (1 + \hat{n})^2 + (1 + \hat{n}) = \hat{n}^2 + 2\hat{n} + 1 + 1 + \hat{n} \\ &= \hat{n}^2 + 3\hat{n} + 2 \end{aligned}$$

$$(2\hat{a}^+ \hat{a} + 1)^2 = 4\hat{n}^2 + 4\hat{n} + 1$$

$$(\hat{a}^+ + \hat{a})^4 = 6\hat{n}^2 + 6\hat{n} + 3 + \dots = 3(1 + 2\hat{n} + 2\hat{n}^2) + \dots$$

where $\langle n^0 | \dots | n^0 \rangle = 0$

$$\Rightarrow \underline{E_n^1} = \frac{\hbar^2}{4m^2\omega^2} 3\lambda \langle n^0 | 1 + 2\hat{n} + 2\hat{n}^2 | n^0 \rangle = \underline{\frac{3\hbar^2\lambda}{4m^2\omega^2} (1 + 2n + 2n^2)} \checkmark$$

as desired.

(2) The perturbative expansion breaks down once

$$E_n^1 \gtrsim E_n^0 \quad \text{or} \quad \frac{E_n^1}{E_n^0} \gtrsim 1$$

$$\frac{E_n^1}{E_n^0} = \frac{3\hbar^2 \lambda (1+2n+2n^2)}{4m^2 \omega^2 \cdot (n+\frac{1}{2})\hbar\omega} \gtrsim 1$$

$$\Rightarrow \frac{2m^2 \omega^3}{3\hbar \lambda} \lesssim \frac{1+2n+2n^2}{1+2n} = 1 + \frac{n^2}{n+\frac{1}{2}}$$

For sufficiently large n the right hand side (which asymptotically grow like n) always becomes larger than the left hand side, hence for sufficiently large n , $\frac{E_n^1}{E_n^0}$ is no longer a small perturbation, but large, and the expansion breaks down.

The reason is that for large x , $\lambda x^4 > \frac{m}{2} \omega^2 x^2$, and higher-lying states have wavefunctions extending to larger x -values (their classical turning points grow in magnitude with n). Once this happens, the λx^4 -part of the potential provides a larger contribution to the total energy than the $\frac{m}{2} \omega^2 x^2$ -part, and the expansion around $\frac{m}{2} \omega^2 x^2$ diverges.

Exercise 17.2.2 (p. 457)

$$\begin{aligned}\hat{H}_{\text{spin}} &= -\hat{\vec{\mu}} \cdot \vec{B} = -\gamma \hat{\vec{S}} \cdot \vec{B} = -\frac{\hbar}{2} \gamma \hat{\vec{\sigma}} \cdot \vec{B} \quad (\hat{\vec{S}} = \frac{\hbar}{2} \hat{\vec{\sigma}}) \\ &= -\frac{\hbar}{2} \gamma (B_1 \hat{\sigma}_x + B_0 \hat{\sigma}_z)\end{aligned}$$

This is the total spin Hamiltonian, i.e. the part of the Hamiltonian controlling the spin evolution.

We write

$$\hat{H}_{\text{spin}} = \hat{H}^0 + \hat{H}' = -\frac{\hbar}{2} \gamma B_0 \hat{\sigma}_z - \frac{\hbar}{2} \gamma B_1 \hat{\sigma}_x$$

The unperturbed Hamiltonian has the following eigenstates and energies:

$$\hat{H}^0 |n^0\rangle = E_n^0 |n^0\rangle \Leftrightarrow -\frac{\hbar}{2} \gamma B_0 \hat{\sigma}_z |m^0\rangle = E_m^0 |m^0\rangle$$

where $m = \pm 1 \equiv \pm$

$$\hat{\sigma}_z |+\rangle = |+\rangle$$

$$\hat{\sigma}_z |-\rangle = -|-\rangle$$

$$\Rightarrow \boxed{E_{\pm}^0 = \mp \gamma B_0 \frac{\hbar}{2}}$$

The first-order energy shift is

$$\boxed{E_{\pm}' = \langle \pm^0 | -\frac{\hbar}{2} \gamma B \sigma_x | \pm^0 \rangle = 0} \quad \text{since } \langle +^0 | \hat{\sigma}_x | +^0 \rangle = 0$$
$$= \langle -^0 | \hat{\sigma}_x | -^0 \rangle$$

(expectation value of $\hat{\sigma}_x$ in $\hat{\sigma}_z$ -eigenstates is zero)

The second-order energy shift is

$$E_+^2 = \frac{|\langle -^0 | (-\frac{\hbar}{2} \gamma B_1 \hat{\sigma}_x) | +^0 \rangle|^2}{E_+^0 - E_-^0} = - \frac{\frac{\hbar^2}{4} \gamma^2 B_1^2 |\langle -^0 | \hat{\sigma}_x | +^0 \rangle|^2}{\gamma B_0 \hbar}$$

$$\langle -^0 | \hat{\sigma}_x | +^0 \rangle = (0 \ 1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (0 \ 1) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1$$

$$\boxed{E_+^2 = -\frac{1}{4} \hbar \gamma \frac{B_1^2}{B_0}}$$

$$E_-^2 = \frac{|\langle +^0 | -\frac{\hbar}{2} \gamma B_1 \hat{\sigma}_x | -^0 \rangle|^2}{E_-^0 - E_+^0} = -E_+^2$$

$$\Rightarrow \boxed{E_+^2 = -E_-^2 = -\frac{\hbar}{4} \gamma \frac{B_1^2}{B_0}}$$

The first order shift in the states is

$$|+'\rangle = | -^0 \rangle \frac{\langle -^0 | -\frac{\hbar}{2} \gamma B_1 \hat{\sigma}_x | +^0 \rangle}{E_+^0 - E_-^0} = + \frac{\frac{\hbar}{2} \gamma B_1}{\gamma \hbar B_0} | -^0 \rangle$$

$$\boxed{|+'\rangle = \frac{1}{2} \frac{B_1}{B_0} | -^0 \rangle}$$

and

$$\boxed{|-'\rangle = | +^0 \rangle \frac{\langle +^0 | -\frac{\hbar}{2} \gamma B_1 \hat{\sigma}_x | -^0 \rangle}{E_-^0 - E_+^0} = -\frac{1}{2} \frac{B_1}{B_0} | +^0 \rangle}$$

• The exact eigenstates of

$$\hat{A}_{\text{spin}} = -\frac{\hbar}{2} \gamma \hat{\vec{\sigma}} \cdot \vec{B} = -\frac{\hbar}{2} \gamma B \hat{n} \cdot \hat{\vec{\sigma}} \quad \left(\hat{n} = \frac{\vec{B}}{B} \right)$$

are $|\hat{n}, \pm\rangle$ with $\hat{n} \cdot \hat{\vec{\sigma}} |\hat{n}, \pm\rangle = \pm |\hat{n}, \pm\rangle$

and energy eigenvalues $\boxed{E_{\pm} = \mp \frac{\hbar}{2} \gamma B}$

The energy eigenvalues can be expanded for small $\frac{B_1}{B_0}$ as follows:

$$\begin{aligned}
 E_{\pm} &= \mp \frac{\hbar}{2} \gamma B = \mp \frac{\hbar}{2} \gamma \sqrt{B_0^2 + B_1^2} = \mp \frac{\hbar}{2} \gamma B_0 \sqrt{1 + \frac{B_1^2}{B_0^2}} \\
 &= \underbrace{\mp \frac{\hbar}{2} \gamma B_0}_{E_{\pm}^0} + \underbrace{\mp \frac{\hbar}{2} \gamma B_0 \frac{1}{2} \frac{B_1^2}{B_0^2}}_{E_{\pm}^2 \text{ (second order in } B_1/B_0!)} + \dots \\
 \Rightarrow \boxed{E_{\pm}^2 = \mp \frac{\hbar}{4} \gamma \frac{B_1^2}{B_0}} &\quad \text{which agrees with our perturbative result.}
 \end{aligned}$$

The states can be expanded as follows:

$$\hat{n} = \left(\frac{B_1}{B}, 0, \frac{B_0}{B} \right) \equiv (\sin \vartheta, 0, \cos \vartheta) \quad (\varphi = 0)$$

$$\begin{aligned}
 |\hat{n}, +\rangle &\stackrel{(14.3.28)}{\longleftrightarrow} \begin{pmatrix} \cos \vartheta/2 \\ \sin \vartheta/2 \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{1}{2}(1 + \cos \vartheta)} \\ \sqrt{\frac{1}{2}(1 - \cos \vartheta)} \end{pmatrix} \\
 &= \begin{pmatrix} \sqrt{\frac{1}{2}(1 + \frac{B_0}{B})} \\ \sqrt{\frac{1}{2}(1 - \frac{B_0}{B})} \end{pmatrix}
 \end{aligned}$$

$$\boxed{\frac{B_0}{B} = \frac{B_0}{\sqrt{B_0^2 + B_1^2}} = \frac{1}{\sqrt{1 + \frac{B_1^2}{B_0^2}}} \approx 1 - \frac{1}{2} \frac{B_1^2}{B_0^2}}$$

$$\approx \begin{pmatrix} \sqrt{1 - \frac{1}{4} \frac{B_1^2}{B_0^2}} \\ \sqrt{\frac{1}{4} \frac{B_1^2}{B_0^2}} \end{pmatrix} \approx \begin{pmatrix} 1 \\ \frac{1}{2} \frac{B_1}{B_0} \end{pmatrix} + O\left(\left(\frac{B_1}{B_0}\right)^2\right)$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{2} \frac{B_1}{B_0} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + O\left(\left(\frac{B_1}{B_0}\right)^2\right)$$

$$\begin{aligned}
 \Rightarrow |\hat{n}, +\rangle &= |+\rangle + \frac{1}{2} \frac{B_1}{B_0} |-\rangle + O\left(\left(\frac{B_1}{B_0}\right)^2\right) \checkmark \\
 &= |+\rangle + |+\rangle' + \dots
 \end{aligned}$$

For $|\hat{n}, -\rangle$ we get

$$|\hat{n}, -\rangle \xleftrightarrow{(4.3.28)} \begin{pmatrix} -\sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} -\sqrt{\frac{1}{2}(1-\cos\theta)} \\ \sqrt{\frac{1}{2}(1+\cos\theta)} \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{2} \frac{B_1}{B_0} \\ 1 \end{pmatrix} + O\left(\left(\frac{B_1}{B_0}\right)^2\right) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \frac{1}{2} \frac{B_1}{B_0} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + O\left(\left(\frac{B_1}{B_0}\right)^2\right)$$

$$\Rightarrow \boxed{|\hat{n}, -\rangle = |-^0\rangle - \frac{1}{2} \frac{B_1}{B_0} |+^0\rangle + O\left(\left(\frac{B_1}{B_0}\right)^2\right) \checkmark}$$

$$= |-^0\rangle + |-^1\rangle + \dots$$

The first-order terms of this expansion also agree with the first-order perturbation theory results.

Exercise 17.2.3, p. 457

- (1) A uniformly charged sphere of radius R and total charge e has charge density $\rho = \frac{e}{\frac{4\pi}{3}R^3}$.

We can get the electric field $\vec{E}(r)$ (which points radially due to symmetry) from Gauss' law:

$$\oint_r \vec{E}(r) \cdot d\vec{\alpha} = (4\pi r^2) E(r) = 4\pi \underbrace{\int_0^r r'^2 dr' \rho(r')}_{q(r)} = \begin{cases} \frac{4\pi}{3} r^3 \frac{e}{\frac{4\pi}{3} R^3} & (r < R) \\ e & (r \geq R) \end{cases}$$
$$\Rightarrow E(r) = \begin{cases} \frac{er}{R^3} & (r < R) \\ \frac{e}{r^2} & r \geq R \end{cases}$$

Since $\vec{E} = -\vec{\nabla}\phi(r)$, we get for $r < R$

$$\phi_<(r) - \phi_<(0) = -\int_0^r E(r') dr' = -\frac{er^2}{2R^3}$$

If we normalize $\phi(\infty) = 0$, then we have for $r > R$

$\phi_>(r) = \frac{e}{r}$. We must choose $\phi_<(0)$ such that $\phi_<(R) = \phi_>(R) = \frac{e}{R}$:

$$\frac{e}{R} - \phi_<(0) = -\frac{e}{2R} \Rightarrow \phi_<(0) = \frac{3e}{2R}$$

$$\Rightarrow \phi(r) = \begin{cases} -\frac{3e}{2R} - \frac{er^2}{2R^3} & r \leq R \\ \frac{e}{r} & r > R \end{cases}$$

$$\text{and } \boxed{V(r) = -e\phi(r) = \begin{cases} -\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} & r \leq R \\ -\frac{e^2}{r} & r > R \end{cases}}$$

$$(2) \quad E'_{100} = \langle 100 | \left(-\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r} \right) \theta(R-r) | 100 \rangle$$

$\uparrow$
 only for $r \leq R$ $V(r)$ deviates
 from $-\frac{e^2}{r}$

$$= \frac{1}{\pi a_0^3} \int_0^R 4\pi r^2 dr \underbrace{e^{-2r/a_0}}_{\approx 1 \text{ since } R \ll a_0} \left(-\frac{3e^2}{2R} + \frac{e^2 r^2}{2R^3} + \frac{e^2}{r} \right)$$

$$\left(\xi = \frac{r}{R} \right) \quad \approx \frac{4R^2}{a_0^3} \int_0^1 \xi^2 d\xi \left(-\frac{3}{2}e^2 + \frac{1}{2}e^2 \xi^2 + \frac{e^2}{\xi} \right)$$

$$= \frac{4R^2}{a_0^3} \left(-\frac{3}{2}e^2 \frac{1}{3} + \frac{1}{2}e^2 \frac{1}{5} + e^2 \frac{1}{2} \right) = \frac{2e^2 R^2}{a_0^3} \left(-1 + \frac{1}{5} + 1 \right)$$

$$E'_{100} \approx \frac{2}{5} \frac{e^2 R^2}{a_0^3} \checkmark$$

Exercise 17.2.4, p. 457

(1) Thomas - Reiche - Kuhn sum rule: $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$

$$\sum_{n'} (E_{n'} - E_n) |\langle n' | \hat{x} | n \rangle|^2 =$$

$$\frac{1}{2} \sum_{n'} [\langle n | \hat{x} \hat{H} | n' \rangle \langle n' | \hat{x} | n \rangle - \langle n | \hat{H} \hat{x} | n' \rangle \langle n' | \hat{x} | n \rangle \\ + \langle n | \hat{x} | n' \rangle \langle n' | \hat{H} \hat{x} | n \rangle - \langle n | \hat{x} | n' \rangle \langle n' | \hat{x} \hat{H} | n \rangle]$$

$$= \frac{1}{2} [\langle n | [\hat{x}, \hat{H}] \hat{x} | n \rangle + \langle n | \hat{x} [\hat{H}, \hat{x}] | n \rangle]$$

$$= -\frac{1}{2} \langle n | [[\hat{H}, \hat{x}], \hat{x}] | n \rangle = -\frac{1}{4m} \langle n | [[\hat{p}^2, \hat{x}], \hat{x}] | n \rangle$$

$$[\hat{p}^2, \hat{x}] = \hat{p} [\hat{p}, \hat{x}] + [\hat{p}, \hat{x}] \hat{p} = -2i\hbar \hat{p}$$

$$-2i\hbar [\hat{p}, \hat{x}] = (-2i\hbar)(-i\hbar) = -2\hbar^2$$

$$\Rightarrow \boxed{\sum_{n'} (E_{n'} - E_n) |\langle n' | \hat{x} | n \rangle|^2 = +\frac{\hbar^2}{2m}} \checkmark$$

(2) harmonic oscillator:

$$\sum_{n'} (E_{n'} - E_n) |\langle n' | \hat{x} | n \rangle|^2 = \sum_{n'} \hbar \omega (n' - n) |\langle n' | \hat{x} | n \rangle|^2 =$$

$$|\langle n' | \hat{x} | n \rangle|^2 = \frac{\hbar}{2m\omega} |\langle n' | \hat{a} + \hat{a}^\dagger | n \rangle|^2 = \frac{\hbar}{2m\omega} [\delta_{n', n-1} (\sqrt{n})^2 + \delta_{n', n+1} (\sqrt{n+1})^2]$$

$$= \frac{\hbar}{2m\omega} (n \delta_{n', n-1} + (n+1) \delta_{n', n+1})$$

$$= \frac{\hbar^2}{2m} (-n + (n+1)) = \frac{\hbar^2}{2m} \checkmark$$

Problem 5

Compute $|n^2\rangle$; from (17.2.10), taking the second term of the Taylor expansion of the translation operator (which is $\sim \mathcal{E}^2$):

$$\begin{aligned}
 |n^2\rangle &= -\frac{1}{2} \frac{q^2 \mathcal{E}^2}{m^2 \omega^4 \hbar^2} \hat{p}^2 |n^0\rangle \\
 &= \frac{1}{2} \frac{q^2 \mathcal{E}^2}{m^2 \omega^4 \hbar^2} \frac{\hbar m \omega}{2} (\hat{a} - \hat{a}^\dagger)^2 |n^0\rangle \\
 &= \frac{1}{4 \hbar \omega} \frac{q^2 \mathcal{E}^2}{m \omega^2} (\hat{a}^2 - \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger + \hat{a}^{\dagger 2}) |n^0\rangle \\
 &= \frac{1}{4 \hbar \omega} \frac{q^2 \mathcal{E}^2}{m \omega^2} \left(\sqrt{n(n-1)} |(n-2)^0\rangle - n |n^0\rangle - (n+1) |n^0\rangle \right. \\
 &\quad \left. + \sqrt{(n+1)(n+2)} |(n+2)^0\rangle \right)
 \end{aligned}$$

$$\Rightarrow |n^2\rangle = \frac{1}{4 \hbar \omega} \frac{q^2 \mathcal{E}^2}{m \omega^2} \left(\sqrt{n(n-1)} |(n-2)^0\rangle - (2n+1) |n^0\rangle + \sqrt{(n+1)(n+2)} |(n+2)^0\rangle \right)$$

Now, compute E_n^3 :

Look at λ^3 terms in (17.1.4):

$$\hat{H}^0 |n^3\rangle + \hat{H}^1 |n^2\rangle = E_n^0 |n^3\rangle + E_n^1 |n^2\rangle + E_n^2 |n^1\rangle + E_n^3 |n^0\rangle$$

Project with $\langle n^0 |$:

$$\langle n^0 | \hat{H}^1 |n^2\rangle = \underbrace{E_n^1}_{=0 \text{ (17.2.2)}} \langle n^0 | n^2 \rangle + E_n^2 \underbrace{\langle n^0 | n^1 \rangle}_{=0 \text{ from (17.2.11)}} + E_n^3$$

$$\begin{aligned}
 \Rightarrow E_n^3 = \langle n^0 | \hat{H}^1 |n^2\rangle &= \frac{1}{4 \hbar \omega} \frac{q^2 \mathcal{E}^2}{m \omega^2} \left(\sqrt{n(n-1)} \langle n^0 | -q \mathcal{E} \hat{X} | (n-2)^0 \rangle \right. \\
 &\quad \left. - (2n+1) \langle n^0 | -q \mathcal{E} \hat{X} | n^0 \rangle \right. \\
 &\quad \left. + \sqrt{(n+1)(n+2)} \langle n^0 | -q \mathcal{E} \hat{X} | (n+2)^0 \rangle \right)
 \end{aligned}$$

Since $\langle n^0 | \hat{X} | n'^0 \rangle \neq 0$ only if $n' = n \pm 1$, all three matrix elements in this last expression vanish and thus

$$\boxed{E_n^3 = 0}.$$

Exercise 17.2.7 (p. 463)

$$\hat{H} = \hat{H}^0 + \hat{H}' = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2} \hat{x}^2 - qE\hat{x}$$

Find $\hat{\Omega}$ such that $\hat{H}' = [\hat{\Omega}, \hat{H}^0]$

$$[\hat{p}, \hat{x}^2] = \hat{x} [\hat{p}, \hat{x}] + [\hat{p}, \hat{x}] \hat{x} = -2i\hbar \hat{x}$$

$$\begin{aligned} \Rightarrow \underline{-qE\hat{x}} &= \frac{-qE}{-2i\hbar} [\hat{p}, \hat{x}^2] = \frac{qE}{2i\hbar} \frac{1}{m\omega^2} \left[\hat{p}, \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2} \hat{x}^2 \right] \\ &= \underline{\left[-\frac{iqE}{m\hbar\omega^2} \hat{p}, \hat{H}^0 \right]} \end{aligned}$$

$$\Rightarrow \hat{\Omega} = -\frac{iqE}{m\hbar\omega^2} \hat{p} \text{ does the job}$$

$$\text{Now, } E_n^2 = \sum_m' \frac{|\langle n^0 | \hat{H}' | m^0 \rangle|^2}{E_n^0 - E_m^0} = (17.2.39)$$

$$= \langle n^0 | \hat{H}' \hat{\Omega} | n^0 \rangle - \underbrace{\langle n^0 | \hat{H}' | n^0 \rangle}_{=0} \langle n^0 | \hat{\Omega} | n^0 \rangle$$

$$= \langle n^0 | (-qE\hat{x}) \left(-\frac{iqE}{m\hbar\omega^2} \hat{p} \right) | n^0 \rangle$$

$$= \frac{q^2 E^2}{m\hbar\omega^2} \sqrt{\frac{\hbar}{2m\omega}} \sqrt{\frac{\hbar m\omega}{2}} \underbrace{\langle n^0 | (\hat{a} + \hat{a}^\dagger)(\hat{a} - \hat{a}^\dagger) | n^0 \rangle}_{\langle n^0 | [\hat{a}^\dagger, \hat{a}] | n^0 \rangle = -1}$$

$$= -\frac{q^2 E^2}{2m\omega^2}$$

This agrees with the earlier result (17.2.6).

The earlier calculation involved computing transition matrix elements of $\hat{x}$, this calculation instead evaluated a diagonal matrix element of $i\hat{x}\hat{p}$.